Christos Giannakopoulos

Data Engineer & Computational Scientist
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PROFESSIONAL SUMMARY

Data Engineer and Computational Scientist with a Physics Ph.D. and **5+ years** of hands-on designing, implementing, and monitoring, **production-grade ETL pipelines** on distributed HPC clusters processing multi-terabyte datasets. Deep expertise in Python (5+ years) and MATLAB (6+ years) and a proven track record of converting research prototypes into versioned, reproducible analytical systems adopted as analysis standards by a 50+ person collaboration. Proficient in developing sophisticated data modeling schemas for data collection, analysis and validation, by using advanced mathematical, statistical, and machine learning tools.

TECHNICAL SKILLS

Programming: Python (5+ yrs), SQL (2+ yrs), MATLAB (6+ years), Bash
Data Engineering: scikit-learn, TensorFlow, Keras, NumPy, Pandas, SciPy, Matplotlib, supervised & unsupervised ML, feature engineering
NLP: spaCy, KeyBERT/BERTopic, LLM fine-tuning, Agentic RAG, Bayesian Inference, Gaussian Processes
Distributed / Big Data: HPC/SLURM clusters (Harvard Cannon, multi-terabyte pipelines), job scheduling, distributed compute management
MLOps & Dev Tools: Git/GitHub, Docker, Linux, end-to-end pipeline orchestration, Docker

EXPERIENCE

Research Data Scientist — *University of Cincinnati* 2021 – 2026

Ph.D. student researcher embedded in a multi-institution, federally funded physics program. Owned analytics and codebase infrastructure for a 50+ person collaboration with deliverables reviewed by program leadership.

- **Designed, implemented, and monitored production ETL pipelines** processing **multi-terabyte datasets** on distributed HPC/SLURM clusters (including Harvard's Cannon HPC), with version-controlled outputs adopted as the **program-wide analytical standard** for a 50+ person multi-institution collaboration.
- **Redesigned core pipeline architecture**, improving runtime by **8×** and enabling near real-time data quality monitoring across distributed compute infrastructure.
- **Developed model validation frameworks** to quantify previously unmeasured systematic uncertainties and benchmark end-to-end system performance, informing hardware design trade-offs and achieving a **78% reduction in compute costs**.
- Built **statistical data quality and provenance tools** that cross-validated pipeline outputs against independent measurements, enabling data-driven decisions for program stakeholders and meeting analysis deadlines **50% sooner**.
- **Optimized predictive models** on complex, multi-modal datasets (time series, point clouds, spatial maps) using Gaussian processes and Bayesian inference, improving detection sensitivity by 2×.
- Built an **end-to-end NLP data pipeline** using spaCy to ingest and process large-scale, multi-source unstructured data, enabling named entity recognition and topic detection, reducing manual review time by 35%.
- **Architected a self-correcting RAG agent** leveraging the 8B Llama 3.1 LLM, integrating a domain-specific knowledge base with a retrieval pipeline for automated Q&A.
- Conducted extensive **feature engineering and dimensionality reduction** (PCA/SVD) on high-dimensional sensor datasets, extracting insights that improved model confidence by **95%**.
- **Led a cross-functional team** on a six-month project delivering published analytical outputs reviewed and approved by program leadership as the collaboration standard.
- **Authored publications** establishing the mathematical and computational framework for novel analysis adopted across the program.
- Mentored junior researchers in pipeline best practices, statistical methods, and reproducible, version-controlled analytical workflows.

EDUCATION

Ph.D. in Physics — University of Cincinnati 2026
B.S. in Physics — Hillsdale College 2019

CERTIFICATIONS & TRAINING

Machine Learning Specialization — DeepLearning.AI
3-course specialization covering Neural Networks, CNNs, Random Forests, Reinforcement Learning, Anomaly Detection, Clustering, and Recommender Systems.

Experimental Systems Engineering — Harvard University & Caltech
Designed and characterized precision measurement systems; performed end-to-end hardware validation and statistical error analysis for a next-generation sensor array.

AWARDS & PUBLICATIONS

"**Calibration Measurements of the BICEP3 and BICEP Array CMB Polarimeters from 2017 to 2024**" Christos Giannakopoulos — SPIE — 2024 *(predictive modeling, data analysis, statistical analysis, data mining)*

Best Recitation Instructor — Exceptional Teaching Award University of Cincinnati *2019, 2020*